

Video Depth Anything: Reproduction and Evaluation on Out-of-Domain Videos

Antonio Lucas da Silva Vale^{*}, Jeanderson da Silva Campos^{*},
Lucas Martins Campos Matos^{*} and Haroldo Gomes Barroso Filho[†]

^{*}Computer Engineering Program

[†]Computer Engineering Coordination

Center for Exact and Technology Sciences (CCET)

Federal University of Maranhão (UFMA), São Luís, Maranhão, Brazil

Corresponding author: lucas.mcm@discente.ufma.br

Abstract—Depth estimation from monocular video is central to computer vision, with applications in augmented reality, robotics, and video editing. Single-image models such as Depth Anything V2 generalize well across domains but produce temporal inconsistency, or *flickering*, when applied frame by frame. Video Depth Anything (VDA) addresses this with a lightweight spatiotemporal decoding head trained with a *Temporal Gradient Matching* loss, combined with a sliding-window inference strategy that handles videos of arbitrary length without accumulating scale error. This work reproduces the VDA architecture and inference strategy and evaluates it in a *zero-shot* setting on two datasets outside its original training domain: TUM RGB-D, which provides real ground-truth depth, and a video of a Brazilian cultural celebration, used as a qualitative case study without reference depth. VDA is compared against two competing models originally evaluated by its authors, DepthCrafter and ChronoDepth, under the AbsRel, RMSE, δ_1 , and TAE metrics. Results show equivalent geometric performance between VDA and DepthCrafter, while ChronoDepth achieves greater temporal consistency at the cost of lower geometric accuracy.

I. INTRODUCTION

Monocular depth estimation infers scene distances from a single camera, without specialized sensors. Foundation models such as MiDaS [7] and Depth Anything V2 [2] train on large heterogeneous datasets, both real and synthetic, producing relative depth estimates that generalize across diverse domains. Applied frame by frame to a video, however, these models generate inconsistent depth for the same scene region across consecutive frames, a phenomenon known as *flickering*.

Two lines of work address this. The first adds temporal smoothing modules on top of pretrained image models, as in NVDS [6], which stabilizes predictions using an Optical-Flow-based Warping (OPW) loss. The second recasts the task as conditional generation via video diffusion models, as in DepthCrafter [3] and ChronoDepth [4], conditioning depth on a window of frames simultaneously. These methods improve temporal consistency but inherit from diffusion a high inference cost and, in most cases, a hard limit on the number of frames processable at once.

Video Depth Anything (VDA) [1], the method reproduced here, avoids diffusion entirely. It uses a lightweight spatiotemporal head trained with the *Temporal Gradient Matching* (TGM) loss, which enforces temporal consistency without

requiring optical flow, combined with a keyframe-anchored sliding-window inference strategy that processes arbitrarily long videos with low latency.

This work reproduces the VDA architecture and inference pipeline, without retraining, using the published weights, and evaluates the model **zero-shot** on data outside its original training and evaluation protocols. The evaluation replaces the original benchmark datasets with: (i) TUM RGB-D [11], which provides real sensor depth and enables quantitative comparison; and (ii) a video of Bumba Meu Boi, a popular cultural celebration from Maranhão, used as a qualitative case study in a domain underrepresented in the depth estimation literature, characterized by moving crowds, artificial nighttime lighting, and costumes with complex textures. VDA is compared against DepthCrafter and ChronoDepth using the same metrics as the original paper.

The main contributions are: (i) reproduction of the VDA architecture, mathematical formulation, and inference strategy, validated on out-of-domain data; (ii) zero-shot quantitative evaluation comparing VDA, DepthCrafter, and ChronoDepth on three TUM RGB-D sequences under the original metrics protocol, including identification of ChronoDepth’s inverted disparity convention and its impact on scale alignment; and (iii) qualitative evaluation on a Brazilian cultural video from an underrepresented nighttime domain, contrasting geometric sharpness against temporal stability across all three models.

The paper is organized as follows: Section II reviews related work; Section III covers the mathematical formulation, architecture, and inference strategy of VDA; Section IV describes the datasets and evaluation protocol; Section V reports quantitative and qualitative results; and Section VI closes with conclusions and future work.

II. RELATED WORK

A. Single-Image Depth Estimation

MiDaS [7] established mixed-dataset training for monocular depth, combining diverse real and synthetic sources under an affine-invariant loss that decouples predictions from each dataset’s scale and offset. Depth Anything V2 [2] scaled this approach using DINOv2 [8] as a frozen visual encoder, producing a foundation model that transfers across indoor,

outdoor, and synthetic domains without fine-tuning. Both models deliver high per-frame accuracy but treat each frame independently, which produces flickering when applied to video.

B. Temporal Consistency in Video Depth

NVDS [6] adds a temporal stabilization network on top of a pretrained single-image model. The network takes consecutive frame predictions and warps the previous prediction to the current frame via optical flow, penalizing differences between the warped and current prediction with the OPW loss. This approach requires optical flow at training time and assumes scene points move smoothly between frames, an assumption that breaks at occlusion boundaries and during fast object motion.

DepthCrafter [3] and ChronoDepth [4] treat temporal consistency differently, using video diffusion priors from Stable Video Diffusion [9]. By conditioning the denoising process on a window of frames simultaneously, they produce temporally coherent predictions without explicit optical flow. The cost is inference time: each window requires multiple denoising steps, making them significantly slower than deterministic models. DepthCrafter additionally imposes a hard limit of 110 frames per inference call, which constrains its use on longer videos.

DepthAnyVideo [5] combines synthetic data generation with a diffusion backbone, achieving strong zero-shot generalization at the cost of higher GPU memory requirements than DepthCrafter or ChronoDepth.

VDA [1] occupies a different position: it preserves the fast deterministic inference of single-image models while adding temporal attention layers that enforce consistency without optical flow or diffusion. The TGM loss directly supervises the temporal gradient of disparity, making the model robust to object motion without requiring dense correspondence between frames.

III. THEORETICAL BACKGROUND

A. Problem Formulation and Affine-Invariant Loss

VDA builds on Depth Anything V2 [2], a relative depth model trained on multiple heterogeneous datasets. Because different datasets represent depth at different scales and offsets, the model predicts **disparity** rather than absolute depth. For each pixel, disparity is defined as

$$d = \frac{1}{t}, \quad (1)$$

where t is the metric depth of the point. Working in disparity space normalizes the representation: far points approach zero, close points produce larger values, stabilizing joint training across heterogeneous datasets.

To train across datasets with different scales, the model uses an affine-invariant loss over the full image (HW pixels):

$$\mathcal{L}_l = \frac{1}{HW} \sum_{i=1}^{HW} \rho(d_i^*, d_i), \quad (2)$$

where d_i^* and d_i are the model prediction and ground-truth disparity at pixel i , respectively. The term ρ is the affine-invariant mean absolute error:

$$\rho(d_i^*, d_i) = \left| \hat{d}_i^* - \hat{d}_i \right|, \quad (3)$$

with $\hat{d}_i^*$ and $\hat{d}_i$ denoting the scale-and-shift aligned versions of prediction and ground truth:

$$\hat{d}_i = \frac{d_i - t(d)}{s(d)}, \quad (4)$$

where the alignment functions are

$$t(d) = \text{median}(d), \quad s(d) = \frac{1}{HW} \sum_{i=1}^{HW} |d_i - t(d)|. \quad (5)$$

$t(d)$ centers the disparity distribution, and $s(d)$ normalizes its spread, making the comparison between prediction and ground truth independent of each dataset's scale and offset. This formulation, inherited from MiDaS [7], is the base on which VDA builds its temporal extension, and corresponds to the $\mathcal{L}_{ssi}$ term in Equation (9).

B. Spatiotemporal Architecture

VDA keeps the DINOv2 [8] encoder from Depth Anything V2 **frozen** during training; only the decoding head is updated. Input, whether a video of N frames or a single image ($N=1$), is reshaped into a tensor $X \in \mathbb{R}^{(B \times N) \times C \times H \times W}$: the temporal dimension is merged into the batch dimension before the encoder, preserving compatibility with static-image training.

The key structural change is the *Spatiotemporal Head* (STH), which replaces the standard DPT head, as illustrated in Figure 1. At selected low-resolution positions in the feature pyramid, temporal layers are inserted: each performs multi-head attention along the temporal axis, comparing the same spatial position across the N frames, followed by a feed-forward network, with absolute positional embeddings encoding each frame's position within the window. This lets the network directly compare the same spatial region across frames without the quadratic cost of full spatiotemporal attention over all pixels.

C. Temporal Gradient Matching Loss

Earlier temporal consistency models, such as NVDS [6], assume that depth at corresponding positions between adjacent frames identified via optical flow should remain approximately constant, giving rise to the OPW loss:

$$\mathcal{L}_{OPW} = \frac{1}{N-1} \sum_{i=2}^N \|p_i - \hat{p}_i\|_1, \quad (6)$$

where p_i and p_{i+1} are consecutive depth predictions and $\hat{p}_i$ is the map p_{i+1} warped to frame i via the optical flow transformation. This assumption breaks down in scenes with relative motion between camera and objects, since the depth of a corresponding point can legitimately change as an object moves closer to the camera.

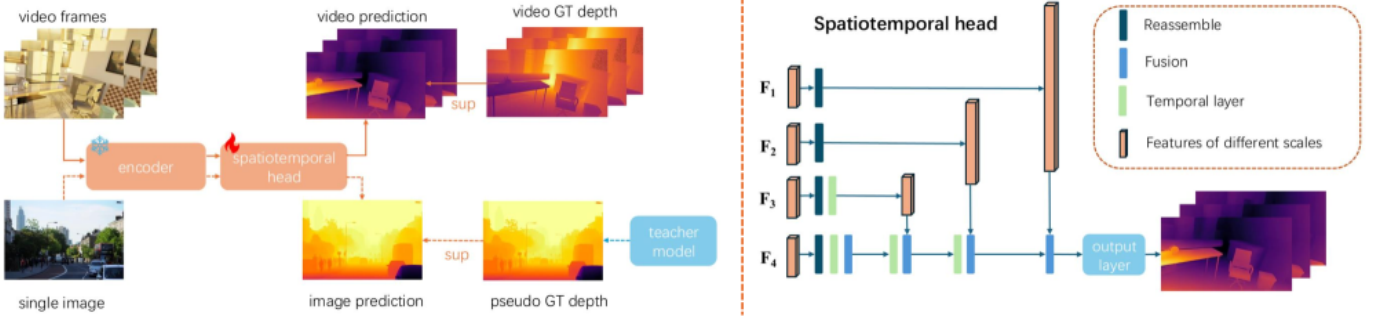


Fig. 1. Overview of the Video Depth Anything architecture. Figure adapted from [1].

An intermediate step replaces direct depth comparison with comparison of the *error* between prediction and ground truth over time, defining the Stable Error (SE):

$$\mathcal{L}_{SE} = \frac{1}{N-1} \sum_{i=1}^{N-1} \left\| |\hat{d}_i - d_i| - |\hat{g}_i - g_i| \right\|_1, \quad (7)$$

where d_i, g_i are the scale-and-shift aligned prediction and ground truth (Eq. (4)), and $\hat{d}_i, \hat{g}_i$ are their versions warped by optical flow between consecutive frames.

VDA's central contribution, the *Temporal Gradient Matching Loss*, removes the optical flow dependency by directly comparing the temporal depth gradient, that is, the change in depth between consecutive frames at the same image coordinates, without any motion correspondence:

$$\mathcal{L}_{TGM} = \frac{1}{N-1} \sum_{i=1}^{N-1} \left\| |d_{i+1} - d_i| - |g_{i+1} - g_i| \right\|_1. \quad (8)$$

In practice, the term is computed only where the ground-truth variation is small ($|g_{i+1} - g_i| < 0.05$), reducing the influence of occlusion boundaries and fast-moving objects that could destabilize training.

The total loss combines the temporal term with the affine-invariant loss of Eq. (2), applied frame by frame:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{TGM} + \beta \mathcal{L}_{ssi}, \quad (9)$$

where α and β balance temporal consistency and per-frame structural quality.

D. Inference Strategy for Long Sequences

The model trains on short windows of $N=32$ frames, but inference must support arbitrarily long videos. VDA handles this in two steps, illustrated in Figure 2.

Keyframe-anchored windows. The video is processed in sliding windows with T_o overlapping frames between consecutive windows. Rather than treating each window independently, a fraction of the input frames for each new window is replaced by the depth already estimated from keyframes (T_k) of the previous window, sampled at regular intervals Δ_k and always including the first frame of the video. This anchors the new window's scale to the global scale already established, reducing the scale drift that accumulates when each window is processed in isolation.

Overlap interpolation. To avoid abrupt transitions at window boundaries, depth in the overlapping frames is blended by linear interpolation between the previous and current window estimates:

$$D_{o_i} = D_{o_i}^{prev} \cdot w_i + D_{o_i}^{cur} \cdot (1 - w_i), \quad (10)$$

with weight w_i decaying linearly from 1 to 0 across the T_o overlap frames.

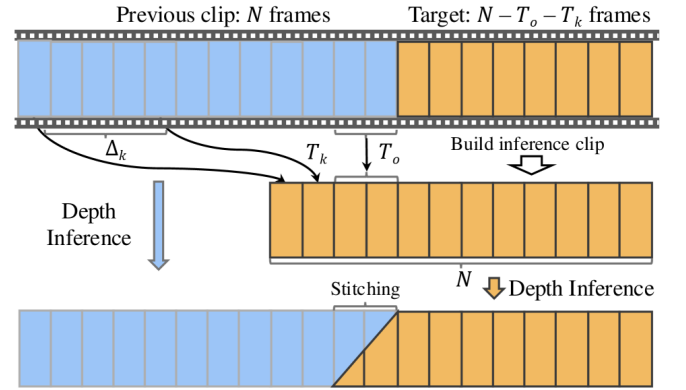


Fig. 2. Keyframe-anchored sliding-window inference strategy. Figure adapted from [1].

IV. METHODOLOGY

A. Datasets and Zero-Shot Evaluation Protocol

Replacing the original benchmark datasets (KITTI, Sintel, ScanNet, NYUv2, and Bonn), two datasets are used here that were not part of any model's training, establishing a **zero-shot** evaluation:

TUM RGB-D [11] is widely used for SLAM and visual odometry evaluation, with real ground-truth depth captured by an RGB-D sensor. Three sequences were selected: *freiburg1_desk*, *freiburg2_desk*, and *freiburg3_long_office_household*. The freiburg1 and freiburg2 sequences are captured with a handheld camera over a desk scene; freiburg3 covers a larger office environment with longer camera traversals and more background clutter. Each sequence is limited to 110 frames, matching the window size used by DepthCrafter in its original protocol [3]. This frame

count also keeps CPU inference tractable for reproduction without GPU access.

Bumba Meu Boi Raízes do Maranhão is a video of a popular cultural celebration from Maranhão, with no available reference depth. It serves as a qualitative case study in a domain largely absent from the depth estimation literature: moving crowds, artificial nighttime lighting, and costumes with complex textures that confound texture-based depth cues.

B. Compared Models

VDA was reproduced in its *small* variant (ViT-S, official weights) and compared against two models from the original paper: **DepthCrafter** [3] and **ChronoDepth** [4], both built on *Stable Video Diffusion* [9]. A third competitor, DepthAnyVideo [5], was tested but excluded from quantitative comparison because it required more GPU memory than available.

C. Video Preprocessing and RGB-Depth Association

TUM RGB-D stores RGB frames and depth maps as separate image sequences, each with its own set of Unix timestamps recorded in plain-text files. Since RGB and depth sensors operate at different rates, 30 fps and 25 fps respectively, the two streams are not aligned by index. For each RGB frame, the closest depth frame by timestamp was selected, rejecting pairs with a time gap above 20 ms to avoid associating frames across scene motion. This produced 110 valid pairs per sequence.

The selected RGB frames were assembled into a video via `ffmpeg` and reencoded in H.264 baseline profile for compatibility with the inference pipelines of all three models. Depth maps were stored as 16-bit PNG files and converted to metric depth in meters by dividing by the TUM scale factor of 5000, following the dataset specification.

D. Disparity Conversion and Scale Alignment

All three models predict relative disparity, but with different normalization conventions. VDA and DepthCrafter follow Eq. (1) directly, $t^* = 1/d^*$, with DepthCrafter normalizing d^* to $[0, 1]$ before inversion. ChronoDepth uses the inverse convention:

$$t^* = \frac{1}{1 - d^*}, \quad (11)$$

identified empirically from implausible AbsRel values under direct inversion. Under direct inversion, pixels with d^* near 1.0, corresponding to nearby objects, map to very large depth values instead of small ones, reversing the near/far ordering and inflating AbsRel beyond 1.0. Equation (11) corrects this, producing AbsRel values consistent with the other two models.

Each prediction t^* is then aligned to the ground truth t by median ratio, analogously to Eq. (5):

$$\lambda = \frac{\text{median}(t)}{\text{median}(t^*)}, \quad \hat{t}^* = \lambda \cdot t^*. \quad (12)$$

Metrics are computed only over valid pixels, defined by mask M where both ground truth and aligned prediction fall in $[0.1 \text{ m}, 10 \text{ m}]$, the range compatible with the TUM RGB-D sensor.

E. Evaluation Metrics

Following the original paper’s protocol, and using the converted and aligned depth $\hat{t}^*$ (Eq. (12)) and validity mask M , the following metrics are computed per sequence over valid pixels:

$$\text{AbsRel} = \frac{1}{|M|} \sum_{i \in M} \frac{|\hat{t}_i^* - t_i|}{t_i}, \quad (13)$$

$$\text{RMSE} = \sqrt{\frac{1}{|M|} \sum_{i \in M} (\hat{t}_i^* - t_i)^2}, \quad (14)$$

$$\delta_1 = \frac{1}{|M|} \sum_{i \in M} \mathbb{I} \left[\max \left(\frac{\hat{t}_i^*}{t_i}, \frac{t_i}{\hat{t}_i^*} \right) < 1.25 \right], \quad (15)$$

where $\mathbb{I}[\cdot]$ is the indicator function. AbsRel and RMSE penalize absolute error, with RMSE more sensitive to large deviations; δ_1 measures the fraction of pixels whose prediction-to-ground-truth ratio stays below 1.25, tolerating small deviations.

The **TAE** (*Temporal Alignment Error*) metric requires no reference depth and can be computed even for the Bumba Meu Boi video. It follows the same structure as the OPW loss of Eq. (6), but between consecutive model predictions rather than between prediction and ground truth:

$$\text{TAE} = \frac{1}{N-1} \sum_{i=1}^{N-1} \|\hat{t}_{i+1}^* - W_{i \rightarrow i+1}(\hat{t}_i^*)\|_1, \quad (16)$$

where $W_{i \rightarrow i+1}(\cdot)$ warps the prediction of frame i to the coordinates of frame $i+1$ via optical flow. Lower TAE values indicate temporally stable predictions.

F. Execution Environment

All experiments ran on Google Colab with an NVIDIA T4 GPU (15 GB). VDA was executed from the official repository with no changes to architecture or inference hyperparameters. DepthCrafter and ChronoDepth were run from their respective official repositories with the default parameters recommended by their authors for memory-constrained GPUs.

V. RESULTS AND DISCUSSION

A. Quantitative Results

Table I reports average results across the three TUM RGB-D sequences for all three models, and Table II breaks down results per sequence.

TABLE I
AVERAGE RESULTS ON TUM RGB-D

Model	AbsRel↓	RMSE↓	δ_1 ↑	TAE↓
VDA (ViT-S)	0.0901	0.4836	0.9464	0.0437
DepthCrafter	0.0886	0.3462	0.9457	0.0382
ChronoDepth	0.1342	0.4866	0.8411	0.0244

VDA and DepthCrafter show near-identical geometric performance: AbsRel and δ_1 differ by less than 0.002, with DepthCrafter recording the lower average RMSE. ChronoDepth posts the lowest TAE of the three, indicating stronger temporal consistency, but at the cost of the lowest geometric accuracy.

TABLE II
PER-SEQUENCE RESULTS ON TUM RGB-D

Sequence	Model	AbsRel↓	RMSE↓	δ_1 ↑	TAE↓
freiburg1_desk	VDA	0.1366	0.6960	0.8982	0.0571
	DepthCrafter	0.0914	0.2999	0.9586	0.0385
	ChronoDepth	0.1403	0.5464	0.8748	0.0372
freiburg2_desk	VDA	0.0732	0.4507	0.9599	0.0434
	DepthCrafter	0.1032	0.4202	0.9237	0.0431
	ChronoDepth	0.1779	0.5840	0.7107	0.0189
freiburg3_office	VDA	0.0606	0.3042	0.9812	0.0306
	DepthCrafter	0.0711	0.3185	0.9547	0.0330
	ChronoDepth	0.0843	0.3295	0.9378	0.0170

B. Qualitative Analysis

Figure 3 compares all three models in the same format used by the original paper for *in-the-wild* DAVIS [10] sequences: input frame followed by each model’s colored depth map. Where the original paper selects six distinct DAVIS videos, here each row corresponds to one of the four sequences in this study, the three TUM RGB-D sequences and the Bumba Meu Boi video, covering both the quantitative and purely qualitative domains.

In *freiburg1_desk*, VDA and DepthCrafter produce structurally close depth maps: the monitor separates cleanly from the background, and the desk shows a smooth gradient. ChronoDepth preserves the same general structure, but with less defined contours on smaller desk objects such as books and keyboard, consistent with its higher AbsRel on this sequence (Table II). In *freiburg2_desk*, VDA achieves the best accuracy of the three models, preserving fine structures of the plant in the scene well. ChronoDepth struggles most on this sequence, with weaker depth separation between desk objects and the background, reflected in its lowest AbsRel and δ_1 values. In *freiburg3_office*, VDA reaches the best result across all nine model-sequence combinations, with good preservation of the plant’s fine structure in the background. ChronoDepth, despite its lowest TAE across TUM sequences here, shows the weakest geometric accuracy, blending the transition between desk objects and the background plane.

In the Bumba Meu Boi video, nighttime domain with artificial lighting and a moving crowd, all three models preserve the foreground dancer’s silhouette reasonably well. ChronoDepth shows less perceptible flickering between frames, consistent with its lower average TAE (Table I), but fine edge delineation, such as the accessories in the dancer’s hand, is sharper in VDA and DepthCrafter. This pattern, ChronoDepth more temporally stable but less geometrically precise, holds across all four sequences.

C. Discussion

One result that diverges from the original paper is how close VDA and DepthCrafter score on TUM RGB-D. On KITTI, Sintel, ScanNet, NYUv2, and Bonn, the original paper shows VDA outperforming DepthCrafter by a clear margin. Two factors explain the difference here: the ViT-S variant used in this study is lighter and less expressive than the ViT-L variant reported as the primary model in the original paper; and TUM

RGB-D belongs to neither model’s training or evaluation sets, reinforcing the zero-shot character of this evaluation.

The TAE results deserve particular attention. VDA’s TAE of 0.0437 is low in absolute terms, even though the model was never evaluated on TUM RGB-D sequences during training. This indicates the TGM loss generalizes the temporal smoothness constraint beyond the training distribution, not just to the specific camera motions and scene types seen during training. ChronoDepth’s TAE of 0.0244, the lowest of the three, comes from the video diffusion prior, which by construction conditions each frame’s depth on surrounding frames. The cost is geometric accuracy: ChronoDepth’s AbsRel of 0.1342 is 49% higher than VDA’s 0.0901.

The trade-off between geometric accuracy and temporal smoothness appears consistently across both the quantitative and qualitative domains. Scenes requiring geometric fidelity, such as 3D reconstruction or autonomous navigation, favor VDA or DepthCrafter. Applications sensitive to visual flickering, such as video post-production or augmented reality overlays, may benefit from ChronoDepth’s stronger temporal consistency.

D. Limitations

This study evaluates only the ViT-S variant of VDA, lighter and less expressive than the ViT-L variant reported as the primary model in the original paper. The quantitative evaluation on TUM RGB-D covers three sequences of 110 frames each, following the DepthCrafter protocol [3]; longer sequences or those with more pronounced camera motion could expose differences not captured in this scope. The Bumba Meu Boi evaluation is entirely qualitative, given the absence of reference depth for that domain. Finally, the median-ratio scale alignment used here is a single global scale factor per sequence; pixel-level scale estimation methods could produce different relative rankings between models.

VI. CONCLUSION AND FUTURE WORK

This work reproduced the mathematical formulation, architecture, and inference strategy of Video Depth Anything and evaluated it zero-shot on data outside its original training domain: TUM RGB-D, providing real sensor depth for quantitative comparison, and a Maranhão cultural video as a qualitative case study. VDA and DepthCrafter show equivalent geometric performance across the three TUM RGB-D sequences. ChronoDepth achieves stronger temporal consistency at the cost of lower geometric accuracy, a trade-off that holds in both the quantitative and qualitative domains. The reproduction also identified ChronoDepth’s inverted disparity convention, which requires a modified inversion formula to produce valid depth estimates under median-ratio scale alignment.

Future work could evaluate the ViT-L variant of VDA under the same protocol to check whether the gap with DepthCrafter seen in the original paper reappears on TUM RGB-D. Extending the Bumba Meu Boi analysis with reference depth obtained via structure-from-motion would turn the qualitative case study into a quantitative one. Including

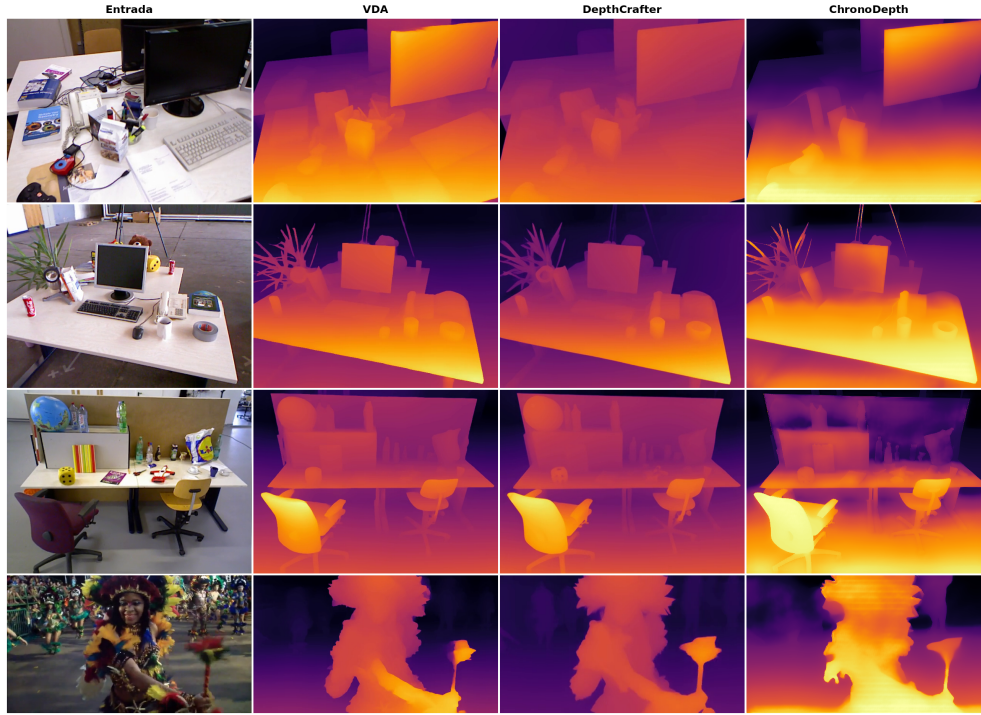


Fig. 3. Qualitative comparison between VDA, DepthCrafter, and ChronoDepth on the four evaluated sequences: freiburg1_desk, freiburg2_desk, freiburg3_office, and Bumba Meu Boi Raízes do Maranhão.

DepthAnyVideo in the comparison, given access to higher-memory GPU hardware, would complete the picture of current diffusion-based methods against VDA on out-of-domain data.

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